

Horticulture Mechanisation and Safety

Science

May, 2026

Horticulture Mechanisation and Safety (A31902)

Short Title: Hort. Mechanisation & Safety
Faculty Engineering
Department Science
Cluster Engineering
Author CDALY
Credits 5

Level: Introductory

Description of Module / Aims

In this module students will learn to operate and maintain horticultural equipment in a safe and efficient way. They will also learn the fundamentals of irrigation from a horticultural perspective.

Programmes

MECH-0038	BSc in Horticulture (WD_SHORT_D)	1 / 2 / M
MECH-0038	BSc in Horticulture (WD_SHORT_D)	2 / 4 / E
MECH-0038	BSc in Horticulture (WD_SHORT_DX)	1 / 2 / M
MECH-0038	BSc in Horticulture (WD_SHORT_DX)	2 / 4 / E

Indicative Content

- Machine shop equipment and practice: hand tools, power tools, wiring 3 pin and 7 pin plugs, crimp connections, junction boxes, and fault finding, Requirements for farm workshop and layout
- Tractor: tractor controls, hitching attachments to three point linkage
- Turf care and grass cutting equipment: principal machines involved in turf care, types of mowers, maintenance and safe use of grass cutting equipment, Aeration equipment for turfgrass
- Irrigation: sources of water, pumps and pipe work, indoor irrigation, outdoor irrigation, calculating irrigation application rates
- Horticulture machinery: operation of horticulture machines, strimmers, hedge cutters, and pedestrian cultivators
- Safe working methods and practices in accordance with associated legislation
- Basic first aid
- Risk assessment and Health and Safety statement for a workplace situation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use measuring and workshop hand and power tools safely and effectively.
2. Operate a range of horticultural equipment.
3. Describe the machines involved in horticulture, e.g. grass cutting and turf care.
4. Describe operation and maintenance of horticultural machines.
5. Describe principles of irrigation.
6. Describe safe working methods and practices in accordance with associated legislation.
7. Complete a risk assessment and Health and Safety statement for a workplace situation.

Learning and Teaching Methods

- Classroom instruction with supporting: videos, models, cutaways and samples.
- Practical demonstrations with follow up practice.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	40%	3,4,5,6
Continuous Assessment	60%	
<i>Practical</i>	50%	1,2
<i>Assignment</i>	10%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrates a limited knowledge of the subject matter. Shows some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Logically and competently carries out practical/professional tasks.

60%-69%: Demonstrates sound knowledge of the subject matter. Demonstrates initiative and the ability when carrying out practical/professional tasks.

70%-100%: Demonstrates imaginative approach, excellent technical ability, awareness, and aptitude in carrying out practical/professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Ryan, M. *_Machinery Workshop Skills_*. Piltown, Kilkenny: Teagasc, 1995.

Supplementary Material

- Bell, B. *_Farm Machinery_*. UK: Farming Press, 1999.
- Bell, B. *_Farm Workshop_*. UK: Farming Press, 1992.
- Bell, B. and S. Cousins. *_Machinery for Horticulture_*. UK: Farming Press, 1997.

Requested Resources

- Lecture Room: Loose Seated